

Feeling old? Here's why.

An in-depth look into your body's ageing process
<http://goo.gl/VSEFu>

Print-a-limb

3D printers of the future will let you print your own limbs
<http://slate.me/wn4SZJ>

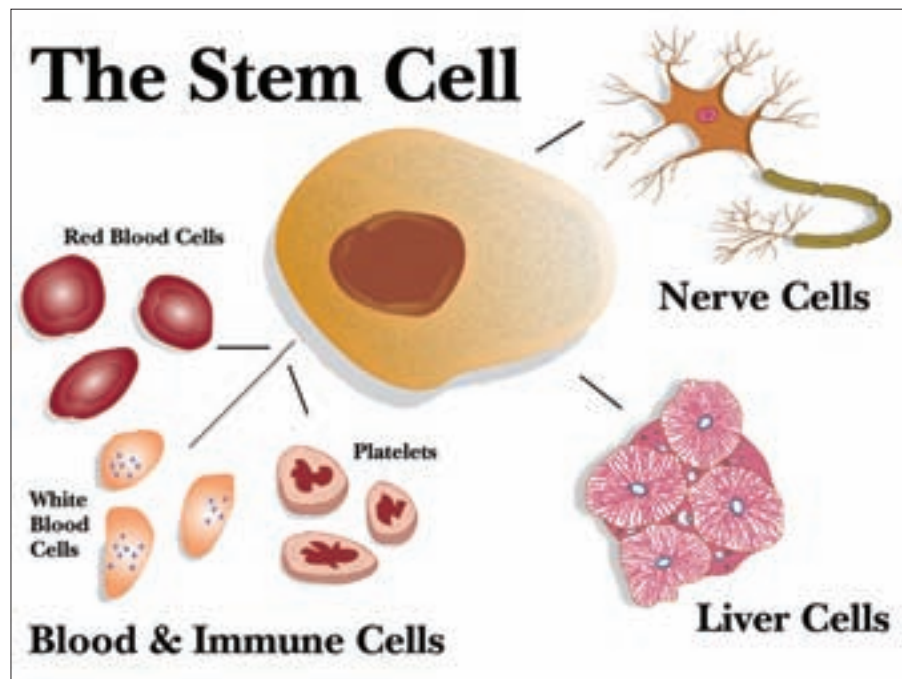
Nano skin

Synthetic skin has for long been produced in laboratories the world over. However, this skin, while it can provide protection, it cannot provide a sense of touch or pressure. Scientists at Stanford university succeeded in producing a type of synthetic skin that can do just that. Consisting of a series of carbon nano-springs, the material can stretch like normal skin and can detect changes in pressure and temperature

Small size = Long life?

Scientists have been experimenting on rats for a considerable amount of time and there is a competition under way for an award known as the "Methuselah Mouse prize" which includes a few million dollars as part of the award. The award will go to the creators of the "ageless" mouse. A lot of researchers are vying for this prize and in the course of this research work that is currently in progress, there are some interesting conclusions that researchers have come to.

It seems that smaller mice live for a relatively longer period than larger ones. It is an interesting observation and the only conclusion that the researchers could come to was that the smaller the mouse, the lower the chance of a failure (organ) at an older stage. The smaller mice were genetically-engineered ones, which lacked a gene required for producing sufficient IGF-1, a hormone that regulates size and is, incidentally, present in all



organisms including us humans. Basically, rats lacking the growth hormone, thyroid stimulating hormone (TSH) and prolactin were found to live longer than their normal brethren. Unfortunately, we can't just stop producing those hormones in our body as they are essential for our survival but it is possible to genetically produce smaller humans. The next time you are ribbed about being tiny, know that you will outlive the ribbers, so rejoice. Bear in mind though that it is not just about being tiny, you need to be tiny and perfect at the same time.

The advantage of being small, in terms of ageing, is statistically very slight (upto a maximum of 10 per cent increase in longevity), but it's a step in the right direction.

Caloric Restriction

Another method to improve your life span is called Caloric Restriction. Research conducted on mice (again) shows that restricting their caloric

intake, while still keeping them well nourished, increased their lifespan by as much as 30 per cent. In humans, that translates to almost 25 years of extra life. Buddhist monks and Indian rishis have been advocating this way of life since time immemorial. Its time we started paying more attention to them.

Stem Cells

The weak spot in this quest for immortality is actually the human body itself. We can theoretically live as long as our brain can. With proper care and maintenance, that can easily be 400-500 years. As mentioned earlier, taking good care of your body helps, but it can only get you through to maybe a 120-130 years and you will be quite frail at that stage. Nowhere near what you would want for immortality. This is where stem cells come in. The magic of stem cells is that these cells can turn into any cell in your body. You might be wondering why this is such a big deal, but don't worry, all will be explained in due course.

Medical science has advanced far enough that you can keep a body on life support even with the absence of an essential organ such as a heart. So, as you grow older, if your organ fails, it is just a matter of replacing it. Torn blood vessels have

